



Elbourne, Paul. 2026. Coordination of predicates and modifiers in a Meaning-Dependent Grammar. *Glossa: a journal of general linguistics* 11(1). pp. 1–39. DOI: <https://doi.org/10.16995/glossa.24182>



Coordination of predicates and modifiers in a Meaning-Dependent Grammar

Paul Elbourne, University of Oxford, paul.elbourne@philosophy.ox.ac.uk

The problem of the coordination of unlike syntactic categories is addressed in the framework of Categoriless Meaning-Dependent Grammar advocated by Elbourne. In particular, the phenomenon of the coordination of predicates and modifiers whose traditional syntactic categories are unlike (*a Republican and proud of it, slowly and with great care*) is attributed to selection being basically for semantic type and never for traditional syntactic categories. Syntactic categories have been abolished in the framework in question. The data in question fall out as an automatic consequence of the more general framework.

Glossa: a journal of general linguistics is a peer-reviewed open access journal published by the Open Library of Humanities. © 2026 The Author(s). This is an open-access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC-BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See <http://creativecommons.org/licenses/by/4.0/>. Articles published by the Open Library of Humanities may also contain other CC-licensed material, or material used or reproduced under an exception or limitation to copyright. Such material is not subject to the article's license. It is either marked with its own license, or marked that it has been reproduced with permission or under fair-use.

 OPEN ACCESS



1 Introduction

This paper addresses a puzzle in the theory of coordination by adopting a version of my (2024) grammar in which syntactic categories are abolished, with their functions being taken over largely by semantic types. The puzzle is the coordination of predicates or modifiers whose traditional syntactic categories are different, as in *a Republican and proud of it* (“DP and AP”) and *slowly and with great care* (“AdvP and PP”). The solution offered by the kind of system that I envisage is that selection is based not on syntactic category but on semantic type and that the coordinated constituents in these examples have the same semantic types.

An expectation that coordinated constituents will have the same syntactic category can arguably be traced back to Chomsky’s *Syntactic Structures*. Speaking of sentences of the form $Z + X + W$ and $Z + Y + W$, where “+” indicates linear adjacency, Chomsky (1957: 36) says that “if X and Y are both constituents, but are constituents of different kinds [...], then we cannot in general form a new sentence by conjunction” (that is to say, of the form $Z + X + \text{and} + Y + W$). On the same page, Chomsky gives a rule that forms conjunctions from constituents that *are* of the same category. Admittedly, this rule gives only a sufficient condition for grammatical conjunctions, not a necessary one, and Chomsky’s initial declaration is carefully qualified (“in general”). But some classic treatments of coordination use only syntactic rules that coordinate phrases of the same syntactic category (Montague 1973; Gazdar 1980; Partee & Rooth 1983). Given that things are in fact more complicated, there is naturally a question as to what an appropriate rule, or set of rules, for coordination actually is.

Counterexamples to Chomsky’s almost-generalization come in several kinds in the literature. This article will offer an analysis of the following two; brief comments on some of the others will be made in the Conclusion. First, there are conjoined predicates of the kind that can come in predicative position after the copula, as in (1) (Sag et al. 1985: 117).

- (1) Pat is a Republican and proud of it. (“DP and AP”)

These also come after other verbs (Dik 1968: 28) and in adjunct position (Peterson 2004: 650):

- (2) He felt quite happy and at ease in his new surroundings. (“AP and PP”)
 (3) John ran down the path, a marked man and desperately afraid. (“DP and AP”)

The second kind of counterexample with which I will deal consists of conjoined modifiers of VPs (Sag et al. 1985: 140):

- (4) We walked slowly and with great care. (“AdvP and PP”)

My basic thesis can be summarized as follows: the conjuncts in these examples are of the same semantic types (an intensional variant of $\langle \text{et}, \text{et} \rangle$ for the predicate cases and a similar variant of

$\langle st, st \rangle$ for the VP modifiers, where s is the type of events); and the word *and* has two homophones (probably among others) that will take phrases of these types and map them to another phrase of the same type. Selection in these cases, then, will be controlled by semantic type and not in the least by syntactic categories, which have been abolished in my (2024) Categoriless Meaning-Dependent Grammar. So the mismatch in syntactic categories, which has been seen as a problem by previous theorists, does not even arise in the current framework.¹

The rest of this article is structured as follows. Section 2 lays out a version of the Categoriless Meaning-Dependent Grammar in my article. Section 3 gives more detail on the treatment in that system of prepositions, adjectives, and what I will call *a-phrases*, which is to say DPs whose determiner is *a*, used as predicates.² An account of the basic predicate and modifier coordination data is to be found in section 4. Some prominent previous accounts of the data in question are reviewed in section 5; and section 6 concludes.

2 A categoriless grammar

The following is a version of the Categoriless Meaning-Dependent Grammar advocated in Elbourne (2024) which is partly based on Stabler’s Minimalist Grammars framework (Stabler 1997; 2011; 2013; Elbourne 2016). The presentation is abbreviated compared to the one in Elbourne (2024), which should be consulted for more detail at many points. There are some additions: lexical entries for the copula, the word *and*, and selected prepositions; and a more sophisticated treatment of tense. On the other hand, the grammar is severely simplified in that it does not include any movement rules or traces; and DPs, which are uniformly items of type $\langle et, t \rangle$ in Elbourne (2024), are here represented only by names, which are assumed, for current purposes, to be of type e . This is largely for the sake of simplicity in the present context, since movement and quantification are orthogonal to the issues on which the current article concentrates.³

2.1 Syntax

The grammar will be a *directional minimalist grammar* (i.e. one that allows linear order to be read off from trees and taken into account in the formulation of Merge; Stabler 2011: 635).

¹ As is common in writings on this topic, I will concentrate on examples involving conjunction and no more than two conjuncts. I do not know of any reason to think that disjunction would work differently than conjunction in the relevant respects. As for examples with more than two conjuncts, I suspect that the best way to go will be to say that conjunctions can bear an indefinitely large number of selector features, all but two of which are optional. This suggestion was made by Neeleman et al. (2023: 53), although they are less keen on it than I am.

² I will sometimes use terms like “DP” as informal labels to pick out various kinds of phrases; but they have no official status in the grammar to be described in section 2.

³ For a Categoriless Meaning-Dependent Grammar that is possibly more realistic in these respects and which also contains the treatments of tense and adjectives found in the current article, see Elbourne (2026). We should note, however, that if Kirkpatrick & Lewerentz (2025) are correct, quantifier phrases might turn out to be of type e after all, meaning that the current grammar is less simplified than one might think.

2.1.1 Features

There are three kinds of features: semantic, phonological, and syntactic. The following abbreviations will be adopted: (donkey) is the semantic value of the word *donkey*; a conventional spelling in phonological slashes, “/donkey/”, represents the phonological features; “donkey” alone, without quotation marks, summarizes the semantic and phonological features; a double colon, “::”, separates the different kinds of features of one word in lexical entries and trees.

The syntactic features in the current simplified grammar will be limited to selector features, which is to say features that cause External Merge. A feature E_L indicates that a constituent needs to combine with another constituent to its left via External Merge. Likewise, with the obvious change, for E_R . Most of the time, selector features just appear as “ E_L ” and “ E_R ”, with no indication of what kind of object must be merged to the left or right of the object that bears the feature. The semantics (in particular (10), the Principle of Interpretability) will ensure that things work out as they should. But occasionally it is necessary to include more information. There are two kinds of case. A selector feature might be annotated with a semantic type: an $E_{L,e}$ feature appears on little *v*, for example, as we see in (7). In that case, something of the indicated type must be merged. Secondly, a selector feature might be annotated with an indication that the merged item must itself bear a certain selector feature (and no other): E_{L,E_R} , for example means that the object that bears the feature must select an object that bears an E_R feature (and no other selector feature) and place it to its left. See section 4.2 for details.

Selector features are descendants of the features variously called c-selectional features (Adger 2003: 96), edge features (Chomsky 2008: 139), selector features (Stabler 2011; Elbourne 2016), and trigger features (Collins & Stabler 2016: 62) by other Minimalist authors. But since they do not contain any information about the syntactic category of the item to be merged, they frequently contain less information than previous versions of this kind of feature.

Selector features cannot be interpreted by the semantics and will thus be referred to as *uninterpretable*.

2.1.2 Trees and sentences

Projection, labelling, headedness, and the notion of specifier will not play a role in this grammar as currently formulated.

The role of a generative grammar is seen as being to generate sentences, where a *sentence*, in the current framework, is defined as a structure of type $\langle i, t \rangle$ (that is, a function from time intervals to truth values) that does not contain any uninterpretable features.

The grammar is set up in such a way that any object produced by it in the course of a derivation will contain at most one node that bears uninterpretable syntactic features. This node will be called the *driver*, since the features on it determine the direction of the computation.

2.1.3 Rules

The following abbreviations will be used in the presentation of the rules. $\tau[\alpha]$ denotes a tree whose driver has a sequence of uninterpretable syntactic features whose first element is α . Given a structure $\tau[\alpha]$, τ denotes the result of erasing feature α .

Since we are operating with a directional minimalist grammar, there are two basic cases of External Merge:

(5) *External Merge*

$$\begin{aligned} \text{a. } \text{merge}(\tau_1[E_{R,\beta}], \tau_2) &= \begin{array}{c} \diagup \quad \diagdown \\ \tau_1 \quad \tau_2 \end{array} \\ \text{b. } \text{merge}(\tau_1[E_{L,\beta}], \tau_2) &= \begin{array}{c} \diagup \quad \diagdown \\ \tau_2 \quad \tau_1 \end{array} \end{aligned}$$

As indicated just above, the alternation of $\tau_1[E_{R/L,\beta}]$ on the left-hand side of these equations and bare τ_1 on the right-hand side means that the feature in square brackets, which triggers Merge, is deleted in the course of it. The symbol β is a variable over the possible further specifications mentioned in section 2.1.1, which can impose conditions on External Merge. If β is empty, that is if there is no further specification, then there is nothing more to be said. If β is a semantic type, then τ_2 must of the same type. And if β is a selector feature, then τ_2 must bear that selector feature (and no other) when it is taken as a syntactic argument; but the result of Merge will be to delete it. Once more, see section 4.2 for more on this.

The following condition will also constrain the application of External Merge:

(6) *Argument Interpretability*

The constituent selected for by a selector feature must not bear any uninterpretable syntactic features after it is merged.

This could be achieved by the selected constituent not bearing any uninterpretable syntactic features to begin with or by its bearing one which is selected for and deleted as part of Merge. A similar condition is imposed by Collins & Stabler (2016: 64) on their operation Triggered Merge.

2.1.4 Lexical entries

Entries in the list in (7) have the following form: “phonological features :: semantic features :: syntactic features”. Optional syntactic features are given in angle brackets: $\langle \rangle$. The others are compulsory. In indications of type, *s* is the type of events and *i* is the type of time intervals. The types of the various denotations are written on the right. (This is for convenience’s sake; types are inherent properties of the objects concerned and do not have to be stipulated separately.) The morphemes BE and PAST appear in different phonological guises in different environments;

I have written the names of these morphemes in capital letters and put them in the place of phonological features, for convenience, even though these names are not strictly speaking phonological features. There are issues here, but it is not the purpose of this article to explore them. “ ε ” indicates an empty sequence.

(7)	/John/ :: $j :: \varepsilon$	e
	/Fido/ :: $o :: \varepsilon$	e
	/care/ :: $c :: \varepsilon$	e
	/a/ :: $\lambda f_{\langle e, it \rangle} . \lambda g_{\langle e, it \rangle} . \lambda x . \lambda t . g(x)(t) \ \& \ f(x)(t) :: E_R$	$\langle eit, \langle eit, eit \rangle \rangle$
	/dog/ :: $\lambda x . \lambda t . dog(x)(t) :: \varepsilon$	$\langle e, it \rangle$
	/cute/ :: $\lambda f_{\langle e, it \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ cute(x)(t) :: \langle E_R \rangle$	$\langle eit, eit \rangle$
	/in/ :: $\lambda x . \lambda f_{\langle e, it \rangle} . \lambda y . \lambda t . f(y)(t) \ \& \ in(y)(x)(t) :: E_R \ \langle E_L \rangle$	$\langle e, \langle eit, eit \rangle \rangle$
	/with/ :: $\lambda x . \lambda f_{\langle e, it \rangle} . \lambda y . \lambda t . f(y)(t) \ \& \ with_1(y)(x)(t) :: E_R \ \langle E_L \rangle$	$\langle e, \langle eit, eit \rangle \rangle$
	/walk/ :: $\lambda R_{\langle i, it \rangle} . \lambda e . \lambda t . walking(e) \ \& \ R(CUL(e))(t) :: E_R$	$\langle iit, sit \rangle$
	/inspect/ :: $\lambda R_{\langle i, it \rangle} . \lambda x . \lambda e . \lambda t . inspection(e) \ \& \ Theme(e)(x) \ \& \ R(CUL(e))(t) :: E_R \ E_R$	$\langle iit, \langle e, sit \rangle \rangle$
	BE :: $\lambda R_{\langle i, it \rangle} . \lambda G_{\langle eit, eit \rangle} . \lambda x . \lambda t . \exists t' (R(t')(t) \ \& \ G(\lambda y . \lambda t'' . \top)(x)(t')) :: E_R \ E_R \ E_{L, e}$	$\langle iit, \langle \langle eit, eit \rangle, eit \rangle \rangle$
	PAST :: $< :: \varepsilon$	$\langle i, it \rangle$
	/slowly/ :: $\lambda F_{\langle s, it \rangle} . \lambda e . \lambda t . F(e)(t) \ \& \ slow(e) :: E_R / E_L / \varepsilon$	$\langle sit, sit \rangle$
	/with/ :: $\lambda x . \lambda F_{\langle s, it \rangle} . \lambda e . \lambda t . F(e)(t) \ \& \ with_2(e)(x) :: E_R \ \langle E_L \rangle$	$\langle e, \langle sit, sit \rangle \rangle$
	/and/ :: $\lambda F_{\langle eit, eit \rangle} . \lambda G_{\langle eit, eit \rangle} . \lambda f_{\langle e, it \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ F(\lambda y . \lambda t' . \top)(x)(t) \ \& \ G(\lambda y . \lambda t' . \top)(x)(t) :: E_{R, \gamma} \ E_{L, \gamma} \ \gamma$	$\langle eiteit, \langle eiteit, eiteit \rangle \rangle$
	/and/ :: $\lambda F_{\langle sit, sit \rangle} . \lambda G_{\langle sit, sit \rangle} . \lambda H_{\langle s, it \rangle} . \lambda e . \lambda t . H(e)(t) \ \& \ F(\lambda e' . \lambda t' . \top)(e)(t) \ \& \ G(\lambda e' . \lambda t' . \top)(e)(t) :: E_R \ E_L \ E_L / E_R$	$\langle sitsit, \langle sitsit, sitsit \rangle \rangle$
	$\varepsilon :: \lambda F_{\langle s, it \rangle} . \lambda x . \lambda t . \exists e (F(e)(t) \ \& \ Agent(e)(x)) :: E_R \ E_{L, e}$	$\langle sit, eit \rangle$
	/not/ :: $\lambda p_{\langle i, t \rangle} . \lambda t . \neg p(t) :: \varepsilon$	$\langle it, it \rangle$

Note that different words in this list have different semantic types, with the exception of words that we want to behave the same, like *John* and *Fido*. Quantifiers, which are not featured here, are of type $\langle eit, \langle eit, it \rangle \rangle$ in this system, another unique type. This is a major step towards the goal of abolishing autonomous syntactic categories, since at the lexical level, at least, words of different traditional syntactic categories now have different semantic types, rendering syntactic categories otiose. We will still, of course, have to make sure that no unwelcome duplications of types arise when larger phrases are created in the course of derivations. For example, we do not want it to

be possible to create a structure of type $\langle \text{sit}, \text{sit} \rangle$ that cannot modify verb phrases. I know of no way of proving that this kind of thing does not happen in systems of this sort. We will just have to analyse an increasing number of constructions and see if any problems arise. But we can note that no such duplications seem to arise in the examples given in Elbourne (2024) and Elbourne (2026). I go through another example in the current paper in section 3.1.

Metalinguage expressions like ‘ $\text{dog}(x)(t)$ ’ are to be read ‘ x is a dog at t ’, and so on. Strictly speaking, the meanings of these lexical items are given in a higher-order formal language with λ -abstraction that is itself subject to (for example) model theoretic interpretation.⁴ (So ‘dog’ as it appears in meanings is a constant, of type $\langle e, \text{it} \rangle$, in this language.) But the meanings should be clear enough as they stand.

The symbol “ $<$ ” in the lexical entry of the past tense morpheme designates a function of type $\langle i, \text{it} \rangle$ such that $< (t)(t')$ iff t occurred before t' . For any event e , $\text{CUL}(e)$ is the culmination of e , which is to say the last temporal point at which e obtains.

The symbol “ γ ” in the first lexical entry of *and* is a variable over the three possibilities that obtain when it comes to selector features: E_R , E_L , and the empty sequence ε . The utility of this will be explained in section 4.2.

The penultimate lexical entry is little v .

2.2 Semantics (Heim & Kratzer 1998)

The grammar will contain the following semantic interpretation rules:

(8) *Functional Application*

If δ is a branching node and $\{\zeta, \eta\}$ the set of its daughters, then δ is in the domain of $\llbracket \cdot \rrbracket$ if both ζ and η are and $\llbracket \zeta \rrbracket$ is a function whose domain contains $\llbracket \eta \rrbracket$. In that case, $\llbracket \delta \rrbracket = \llbracket \zeta \rrbracket(\llbracket \eta \rrbracket)$.

(9) *Lexical Terminals*

If κ is a lexical item, then $\llbracket \kappa \rrbracket$ is given by the semantic features of κ .

Since the simple grammar of this article does not deal with quantification, binding, or context sensitivity, there is no rule of Predicate Abstraction (Heim & Kratzer 1998: 114) and no procedure for interpreting pronouns or traces.

The grammar will also make use of the following principle (Heim & Kratzer 1998: 49):

(10) *Principle of Interpretability*

All nodes in a phrase structure tree must be in the domain of the interpretation function $\llbracket \cdot \rrbracket$.

⁴ The basic strategy is taken over from Montague (1973).

The interpretation of individual lexical items is given in (7).

As remarked in Section 2.1.2, the grammar is meant to generate sentences, where a sentence, in the current framework, is a structure of type $\langle i, t \rangle$ that does not contain any uninterpretable features. The denotations of these structures, being functions from time intervals to truth values, will by convention be applied to the time of utterance to obtain a truth value for an utterance on any given occasion.

3 Adjectives, prepositions, and *a*-phrases

As we can see already in examples (1)–(3), many of the examples with which this article will deal involve adjective phrases, prepositional phrases, and what I will call *a*-phrases, which is to say DPs whose determiner is *a* (*a Republican*, *a marked man*), used in a seemingly predicative fashion. In this section I will describe the treatment of these phrases in the current grammar.

3.1 Adjectives

In this grammar, adjectives are functions from noun denotations to noun denotations. This helps, in terms of the current project, to distinguish them from nouns. This semantic analysis seems to have occurred independently and more or less simultaneously to Clark (1970), Lewis (1970), Montague (1970), and Parsons (1970). This is not too surprising, since it is a very natural analysis.⁵ It is, of course, particularly natural when dealing with adjectives in attributive position (when they are sisters of nouns, or sisters of adjective–noun combinations, and so on, recursively): the adjective takes the noun (or adjective–noun combination) as an argument and maps it to something of the same type as a noun; the new object thus created can be modified by a further adjective or taken as an argument by a determiner, as the case may be.

Here are some representative lexical entries, repeated from (7).⁶

(11) /dog/ :: $\lambda x. \lambda t. \text{dog}(x)(t) :: \varepsilon$

(12) /cute/ :: $\lambda f_{\langle e, it \rangle}. \lambda x. \lambda t. f(x)(t) \ \& \ \text{cute}(x)(t) :: \langle E_R \rangle$

⁵ A reviewer points out that this semantic analysis mirrors the N/N syntactic analysis given to adjectives by early Categorical Grammar (Bar-Hillel 1953) and speculates plausibly that the authors just named were influenced in this way.

⁶ Some might wonder why I have not incorporated degrees into my semantics for adjectives, as done by Kennedy (2007) and others. The answer is that in this paper I am only concerned with what one might call the outward-facing semantics of adjectives, with questions, in other words, of how they combine with other words. And for these purposes Kennedy (2007) too, by the addition of his *pos* morpheme, makes ordinary positive adjectives be of a type $\langle \langle e, t \rangle \rangle$ in his case) that does not involve degrees. I see no obstacle to degrees being added to the current theory in future work that applies it to the kinds of questions with which Kennedy is dealing, such as the semantics of comparatives and superlatives.

In the syntax, these will combine with each other easily by External Merge, assuming that the optional feature E_R is present on the adjective:

$$(13) \quad \begin{array}{c} \diagup \quad \diagdown \\ \text{cute} :: \varepsilon \quad \text{dog} :: \varepsilon \end{array}$$

The feature E_R is deleted when Merge takes place, in accordance with (5a). A straightforward calculation shows that the semantic value of this constituent is the following:

$$(14) \quad \lambda x. \lambda t. \text{dog}(x)(t) \ \& \ \text{cute}(x)(t)$$

This is of type $\langle e, it \rangle$, as desired.

But what about predicative position, when an adjective appears after the copula? My denotation for the copula has already been given in (7). Here are some lexical entries repeated from there that will enable us to analyse an example sentence:

$$(15) \quad \begin{array}{ll} \text{a.} & / \text{Fido} / :: o :: \varepsilon \\ \text{b.} & / \text{cute} / :: \lambda f_{\langle e, it \rangle}. \lambda x. \lambda t. f(x)(t) \ \& \ \text{cute}(x)(t) :: \langle E_R \rangle \\ \text{c.} & \text{BE} :: \lambda R_{\langle i, it \rangle}. \lambda G_{\langle eit, eit \rangle}. \lambda x. \lambda t. \exists t' (R(t')(t) \ \& \ G(\lambda y. \lambda t''. \top)(x)(t')) \\ & \quad :: E_R \ E_R \ E_{L, e} \\ \text{d.} & \text{PAST} :: < :: \varepsilon \end{array}$$

Let us analyse (16):

$$(16) \quad \text{Fido was cute.}$$

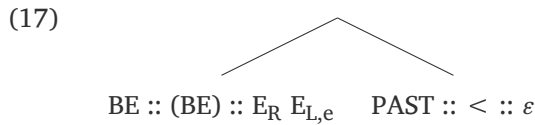
We will consider the derivation in some detail in order to illustrate the working of the current system.

In order to do so, it will be necessary to make a distinction between two ways in which one linguistic constituent might be the argument of another one. One constituent might be the *syntactic argument* of a second: this means that it is selected for by a selector feature on the second. Given (6) (Argument Interpretability), the first constituent in this kind of case will have to bear no uninterpretable syntactic features (unless we are dealing with one of the rare cases in which uninterpretable syntactic features are actually selected for). Alternatively, one constituent might be the *semantic argument* of a second: this means that the denotation of the second is a function that takes as its argument in the semantics the denotation of the first. Given binary branching in the syntax and functional application between sisters in the semantics, if one constituent is the semantic argument of a second, it is its sister. These two ways of being an argument can in principle come apart: for example, in Elbourne (2024), the constituent formerly known as v' (type

$\langle e, t \rangle$) takes as its syntactic argument a generalized quantifier subject (type $\langle e, t \rangle$) that takes it as its argument in the semantics. In the following derivation, I will also need to say that no lexical item in the current grammar and no known constituent constructible from lexical items in the current grammar can take things of a certain type as a semantic argument; in these circumstances I will say that *no known constituent* takes things of this type as a semantic argument.

It is also worth noting ahead of time that (6) (Argument Interpretability) will constrain the derivation by ensuring that any constituent taken as a syntactic argument will have to bear no uninterpretable syntactic features, unless we are dealing with one of the rare cases in which an item *selects* for a certain uninterpretable syntactic feature. The only item in the current fragment that can bear features enabling it to do that is *and*. (Outside the current fragment, I know only of *or*.) Since the current example does not include *and* or *or*, I will assume for current purposes that (6) forbids taking a syntactic argument that has an uninterpretable syntactic feature.

We will start with the copula, whose lexical entry is in (15c). Since it bears an uninterpretable syntactic feature, it cannot be taken as a syntactic argument. In order to combine with something else, then, it will have to use its first selector feature, E_R . The copula is of type $\langle iit, \langle \langle eit, eit \rangle, eit \rangle \rangle$. No known constituent takes things of this type as semantic arguments. So it will have to select for something of type $\langle i, it \rangle$. There is only one thing of this type in our fragment, which is the tense morpheme PAST. So the copula takes this as sister to its right:



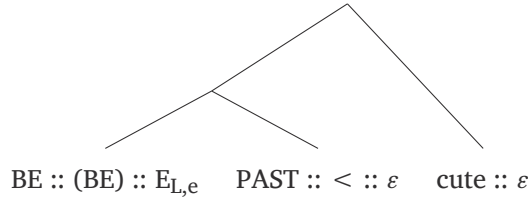
The feature E_R that triggered the Merge is deleted when Merge takes place, in accordance with (5a).

We see that (17) cannot be taken as a syntactic argument, since it contains uninterpretable syntactic features. So it will have to take something as an argument using the next selector feature on its driver, which is E_R . It is of type $\langle \langle eit, eit \rangle, eit \rangle$. No known constituent takes semantic arguments of this type. So it will have to take as a syntactic and semantic argument something of type $\langle eit, eit \rangle$. Now there will turn out to be quite a few things of this type in the current grammar: adjective phrases, prepositional phrases, and *a*-phrases, in fact. We will explore all these options in due course. For now, let us just pick the adjective *cute*.

We will have to use the version of *cute* without its optional feature E_R , since (17) has its own syntactic features. If we left the E_R feature on *cute*, we would violate (6) (Argument Interpretability).⁷ With that duly noted, we perform the Merge and produce the following tree:

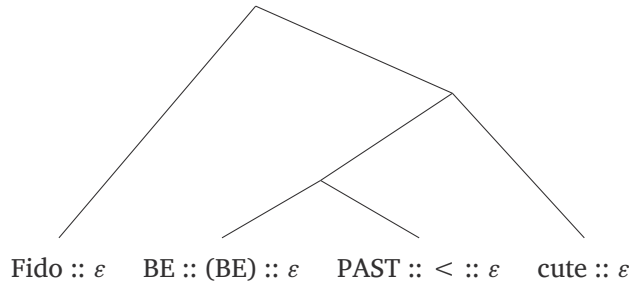
⁷ The optionality of the E_R feature on adjectives represents a change to the system in Elbourne (2024). One result of this change is that little *v* (the penultimate entry in (7)) and BE come with $E_{L,e}$ features to ensure that the right type of

(18)



The E_R feature that triggered Merge is deleted, as usual. This structure contains an uninterpretable syntactic feature, meaning that it cannot be taken as a syntactic argument. This means that the structure will have to take a syntactic argument using its $E_{L,e}$ feature. This mandates that something of type e be taken as syntactic argument at this point. We perform the Merge, picking *Fido* as our type e item, and end up with the following tree:

(19)



This is a sentence by the definition introduced in section 2.1.2, namely a structure of type $\langle i, t \rangle$ that does not contain any uninterpretable features. A straightforward calculation produces the following truth conditions.⁸

$$(20) \quad \lambda t. \exists t' (< (t')(t) \ \& \ \text{cute}(o)(t'))$$

These truth conditions are intuitively adequate. I assume that a post-syntactic process, not dealt with here, will spell out BE PAST as *was*.

I should briefly mention the possibility that constructions like (16) (*Fido was cute*) contain some version of little v . Although little v is perhaps most frequently invoked as a means to add in an Agent, researchers have sometimes suggested that it comes in different flavours. Chomsky (2013: 43, footnote 29), for example, suggests that there are at least two: one flavour would deal with transitive and unergative verbs, while a second one would handle unaccusatives and passives; and other authors such as Folli & Harley (2007) expand the list still more. If there is version of little v in (16), it would presumably take $[[\text{BE PAST}] \text{cute}]$ as its argument. If we assume

thing is merged as the subject; if the E_L features in question did not have an annotation for type, it would be possible to merge an adjective (without an E_R feature) with the structures that little v and BE create, which are of type $\langle e, it \rangle$.

⁸ The only matter to note is that the calculation relies at one point on the principle that $(p \ \& \ T)$, where ' T ' is the truth value True, is equivalent to p . That is the role of the ' T ' in the denotation of BE.

that this flavour of little *v* would be of type $\langle \text{sit}, \text{eit} \rangle$, like the one I list in (7), then presumably [BE PAST] would be of type $\langle \langle \text{eit}, \text{eit} \rangle, \text{sit} \rangle$, with corresponding consequences for BE. This could presumably all be managed, if necessary, but I do not see any need to spell out this alternative any further, for in fact I know of no good evidence to suggest that there is a version of little *v* in (16) and am inclined to conclude that there is no such thing. I am joined in this conclusion, I am pleased to note, by den Dikken (2006: 23–24, 29, 63). I do not know of any author who has argued in detail that constructions like (16) contain little *v*; and Maienborn (2005) argues convincingly that there are no event variables in play in these sentences.

There are many more issues that would ideally be dealt with in a syntactic and semantic analysis of adjectives in the current framework, but I cannot devote time to them now, since they are orthogonal to the focus of the current article. A detailed discussion of this topic can be found in Elbourne (2026).

3.2 Prepositions

Across a wide range of their uses, prepositional phrases have the distribution of adjectives (Heim & Kratzer 1998: 63–65):

- (21) a. Kaline is grey.
b. Kaline is in Texas.
- (22) a. a grey cat
b. a cat in Texas

Looking at these data, the obvious approach in the current context is to say that prepositions are of type $\langle e, \langle \text{eit}, \text{eit} \rangle \rangle$: they first take an internal argument of type *e*; the whole prepositional phrase is of type $\langle \text{eit}, \text{eit} \rangle$, the type of adjectives, which explains their parallel distribution (with the exception of word order with respect to the noun in the attributive case). For *in* this suggests the lexical entry in (7) (Heim & Kratzer 1998: 66), repeated here:

- (23) $/\text{in}/ :: \lambda x. \lambda f_{\langle e, \text{it} \rangle}. \lambda y. \lambda t. f(y)(t) \ \& \ \text{in}(y)(x)(t) :: E_R \ \langle E_L \rangle \qquad \langle e, \langle \text{eit}, \text{eit} \rangle \rangle$

Looking at (22b) in particular, it seems that prepositions come with E_R and E_L features, in that order, so as to put their internal arguments and the nouns they modify in the right places.^{9,10}

⁹ I follow Heim & Kratzer (1998: 65) and others in supposing that a prepositional phrase that follows a noun in English, as in (22b), can be treated as such and not as the only visible manifestation of a reduced relative clause, contrary to what was claimed by Ross (1972: 65).

¹⁰ If prepositions are of type $\langle e, \langle \text{eit}, \text{eit} \rangle \rangle$, this might be thought to create a clash with relational adjectives like *local*, as in *Every sports fan in the country was at a local bar*. I maintain, however, that relational adjectives are of type $\langle \text{eit}, \text{eit} \rangle$ and that the binding in examples like this is brought about by their being context-sensitive in a way that does not increase their arity. Similar analyses have already been proposed by Partee (1989) and Cresswell (1996). See Elbourne (2026) for one detailed way of spelling this out.

We will also have to deal with prepositional phrases in predicative position, as in (21b). Since prepositional phrases like *in Texas* will be of type $\langle \text{eit}, \text{eit} \rangle$, just like adjectives, examples like (21b) will work just like (21a), which in turn will work just like (16), which we have already examined. Since the copula has its own E_R feature to merge in its predicate complements, the E_L feature on prepositions is optional, because of (6) (Argument Interpretability), as in the case of the E_R feature on *cute*.

The above meaning cannot be the only one for prepositions, however, because at least some of them are capable of forming VP-modifiers:

(24) John walked with care.

This suggests the following lexical entry for *with*:

(25) /with/ :: $\lambda x. \lambda F_{\langle s, \text{it} \rangle}. \lambda e. \lambda t. F(e)(t) \ \& \ \text{with}_2(e)(x) :: E_R \ \langle E_L \rangle$ $\langle e, \langle \text{sit}, \text{sit} \rangle \rangle$

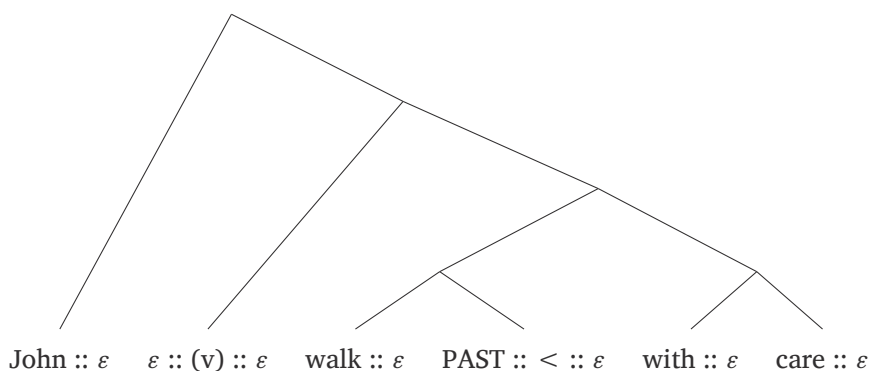
The type $\langle e, \langle \text{sit}, \text{sit} \rangle \rangle$ is obviously parallel to the type $\langle e, \langle \text{eit}, \text{eit} \rangle \rangle$ that *with* has when it occurs outside the verbal realm:

(26) /with/ :: $\lambda x. \lambda f_{\langle e, \text{it} \rangle}. \lambda y. \lambda t. f(y)(t) \ \& \ \text{with}_1(y)(x)(t) :: E_R \ \langle E_L \rangle$ $\langle e, \langle \text{eit}, \text{eit} \rangle \rangle$

The two lexical entries are structurally similar too. This suggests that there may be a systematic process of polysemy at work here, but it would take us too far afield to attempt to uncover it.

Since I have already gone through one derivation in detail, allow me just to state without further justification that the fragment in the current article generates the following structure and truth conditions for (24), assuming for the sake of simplicity that *care* can be treated as a constant (*c*) of type *e*:

(27)



(28) $\lambda t. \exists e (\text{walking}(e) \ \& \ < (\text{CUL}(e))(t) \ \& \ \text{with}_2(e)(c) \ \& \ \text{Agent}(e)(j))$

3.3 α -Phrases

These, as mentioned above, are DPs whose determiner is a , used in predicative fashion. A simple example is a *dog* in (29):

(29) Fido is a dog.

The relevant lexical entry for a was given in (7). It is repeated here:

(30) $/a/ :: \lambda f_{\langle e, it \rangle} . \lambda g_{\langle e, it \rangle} . \lambda x . \lambda t . g(x)(t) \ \& \ f(x)(t) :: E_R \qquad \langle eit, \langle eit, eit \rangle \rangle$

This, it should be stressed, is distinct from the quantificational meaning of a , which is not included in the fragment of this article. The quantificational meaning, in a larger and more comprehensive fragment, would be as follows:

(31) $/a/ :: \lambda f_{\langle e, it \rangle} . \lambda g_{\langle e, it \rangle} . \lambda t . \exists x (f(x)(t) \ \& \ g(x)(t)) :: E_R \qquad \langle eit, \langle eit, it \rangle \rangle$

Ultimately, then, it looks as if the current approach will posit an ambiguity in the word a . But this is very much a standard move in the current context. Heim & Kratzer (1998: 61–63), for example, operate on the assumption that both the copula and the word a are semantically vacuous in (29), meaning that a would have to be ambiguous between vacuity and its normal quantificational meaning.

The meaning in (30) is of type $\langle eit, \langle eit, eit \rangle \rangle$, a unique type in the current system (not to be confused with the type of the eventual quantificational meaning of a , which is $\langle eit, \langle eit, it \rangle \rangle$, or with the suggested semantic type for [BE PAST], which is $\langle \langle eit, eit \rangle, eit \rangle$); it converts noun meanings into meanings of the same type and overall shape as adjective meanings. In particular, assuming that *dog* has the meaning in (32), a *dog* would thereby have the meaning in (33):

(32) $\llbracket \text{dog} \rrbracket = \lambda x . \lambda t . \text{dog}(x)(t)$

(33) $\llbracket a \text{ dog} \rrbracket = \lambda f_{\langle e, it \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ \text{dog}(x)(t)$

(34) $\llbracket \text{cute} \rrbracket = \lambda f_{\langle e, it \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ \text{cute}(x)(t)$

This, it can be seen, is precisely parallel to the meaning of *cute* in (15b), repeated in (34), which means that BE can operate on it as in section 3.1 and produce an appropriate meaning for (29). Meanwhile, note that a *dog*, despite having the meaning in (33), cannot take any nouns (type $\langle e, it \rangle$) as arguments because it does not have any syntactic features to enable it to do so: a in (30) has only one selector feature, an E_R feature that will suffice to merge in a noun; none will be left to enable a *dog* to take any syntactic arguments. It will have to be taken as a syntactic argument by something else, such as the copula.

3.4 Subcategorization requirements of verbs and modifiers

I have been arguing in this section that adjective phrases, prepositional phrases, and *a*-phrases could all be of the same semantic type, namely type $\langle \text{eit}, \text{eit} \rangle$. This will arguably be useful when it comes to explaining the coordination data in section 4.¹¹ But it has been claimed that some English verbs (and some modifiers like *very*) subcategorize for one or two of these traditional syntactic categories but not the others. How can this be if subcategorization does not rely on syntactic categories and does rely on semantic type, as I claim?

This does not seem to be a very widespread phenomenon. Dalrymple (2017: 34) says: ‘There are very few predicates that require a particular c-structure category for their arguments, but a few such predicates are attested. Often, the verbs *wax* and *become* are given as examples.’ In this section, I will describe how the current approach to grammar can deal with the most prominent examples known to me: *seem*, *become*, *make*, *get*, *wax*, and *very*. If there are any others, I trust that the advances made with the current examples will inspire confidence that similar cases will be amenable to analysis in this framework. Briefly, the line that I will be taking is that previous empirical work on these words has not been nearly detailed enough and that it is in fact possible for them to take complements of the allegedly forbidden kinds. The data unearthed strongly suggest that these words take complements that are of type $\langle \text{eit}, \text{eit} \rangle$, to start with, and which are subject to further *semantic* conditions on top of that.

A good place to start is with the verb *seem*, in its use whereby it takes a predicative complement. This verb has sometimes been claimed to subcategorize only for adjective complements, on the basis of data like the following (Maling 1983; Pollard & Sag 1994):

- (35) a. Sandy seems clever.
 b. *Sandy seems out of town.
 c. *Robin seems a cabinet minister.

But, as Maling (1983: 256) showed, this claim is false. If we move away from prepositional phrases whose meanings have to do with physical location and use what Maling calls *metaphorical PPs*, the judgements change:

- (36) a. Lee seems under the weather.
 b. That suggestion seemed completely off the wall.

As for *a*-phrases, the acceptability of (37) contrasts with that of (35c):

¹¹ In a larger fragment, there will be one more kind of $\langle \text{eit}, \text{eit} \rangle$ item, namely restrictive relative clauses. I have not concentrated on those because it is likely that they will come with compulsory E_L features in order to attach them to nouns. They could not be taken as syntactic arguments by the copula or other verbs, then, due to (6) (Argument Interpretability); nor could they appear as free-standing secondary predicates. They cannot appear in the kind of data that we see in (1)–(3); and it seems that we have a good independent explanation of this.

- (37) Robin seems a fool.

Maling (1983: 258) concludes that what is really going on here is that *seem* subcategorizes for a semantic class, the class of *gradable predicates*. The class of gradable predicates cross-cuts the traditional syntactic categories. A semantic analysis of this case has also been endorsed by Pollard & Sag (1994: 104), among others.¹²

Armed with this precedent, let us examine some other verbs that have been claimed to subcategorize for just one or two of adjective phrases, prepositional phrases, and *a*-phrases. Dalrymple (2017: 34), for example, claims that the verb *become* subcategorizes for adjective phrases and *a*-phrases but excludes prepositional phrases, on the basis of data like the following:

- (38) a. Fred became happy.
b. Fred became a professor.
c. *Fred became in the room.

This is incorrect, however. Once again, judgements change when we seek out Maling's class of metaphorical PPs. The following naturally occurring examples (all consulted online on 11 March 2025) are perfectly acceptable:

- (39) a. Today, 22 US states do not have comprehensive laws banning smoking in public. "We have become behind the times", says Erika Sward, the American Lung Association's Assistant Vice President of National Advocacy.¹³
b. Sensors in a device based on the magnetometer—the compass, inclinometer and orientation sensor—can become in need of calibration due to environmental factors.¹⁴
c. When the "Mayflower" turned south, however, it ran into rough, shallow waters and became in danger of tipping over and sinking.¹⁵

We must conclude, then, contrary to previous work, that *become* takes members of all three of the syntactic categories under discussion, but the selection is at least partly governed by a semantic condition. In the terms of the current theory, *become* takes items of $\langle \text{eit}, \text{eit} \rangle$ as complements; but this selection is subject also to a further semantic condition.

A similar story can be told about the verb *make*. Bruening & Al Khalaf (2020: 8) claim that *make* selects for APs and NPs but not PPs, on the basis of data like the following:

¹² Just to be clear, while I do think that a semantic condition will be crucial in analysing the subcategorization behaviour of *seem* (since the complements it takes cross-cut traditional syntactic categories), I wish to remain neutral, for current purposes, about Maling's theory that gradability will be the key.

¹³ [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(14\)61760-8/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(14)61760-8/fulltext).

¹⁴ <https://learn.microsoft.com/en-us/windows/uwp/devices-sensors/sensors>.

¹⁵ <https://www.nps.gov/caco/learn/historyculture/pilgrims.htm>.

- (40) a. I made the child more polite.
 b. I made the child a better host.
 c. *I made the child out of the house.

Again, though, the picture changes when we include metaphorical PPs. The following naturally occurring examples (consulted online on 18 November 2025) are perfectly acceptable:

- (41) By reflecting on the Ten Commandments, Christians can examine their hearts to see where disordered attachments and desires have made them in need of Christ's healing, Pope Francis said Wednesday.¹⁶
- (42) Believe it or not, I have overheard Church of Christ preachers mock the poor who get this aid, without even knowing the particular circumstances which made them in need of such aid.¹⁷
- (43) Considering that Guerra is such a terrific character actor and Willy added a great amount of depth to Buffy's worldbuilding, it's a shame he didn't stick around, but even so the time we got with him made him in need of much more fan appreciation.¹⁸
- (44) His physical injuries weren't life-threatening, but his mental state made him in danger of dying.¹⁹

We must conclude, contrary to previous work, that *make* takes all three of the syntactic categories now under discussion, subject to a semantic constraint; or, in terms of the current theory, that it takes ⟨*ei*,*ei*⟩ items subject to a further semantic constraint.

The word *get* presents a similar picture. In previous work, this verb has been alleged to take adjective phrases and prepositional phrases but not *a*-phrases (Bruening & Al Khalaf 2020: 8):

- (45) a. The sergeant got the guard alert.
 b. The sergeant got the guard on top of the wall.
 c. *The sergeant got the guard a better marksman.

But here too the empirical picture turns out to be more complicated than was previously claimed. An anonymous reviewer for this journal (to whom I am very grateful) points out that the following sentence, with an *a*-phrase as the second internal argument, is perfectly fine:

- (46) The sergeant got the guard a safe distance from the action.

¹⁶ <https://www.catholicnewsagency.com/news/40005/the-ten-commandments-are-gods-x-ray-pope-francis-says>.

¹⁷ <https://biblicalproof.wordpress.com/2015/01/20/why-eating-in-the-church-is-a-sin/>.

¹⁸ <https://whatculture.com/tv/10-most-underrated-buffy-the-vampire-slayer-characters?page=2>.

¹⁹ <https://www.goodreads.com/topic/show/1027026-jamie-s-healing>.

The reviewer comments: ‘So constraints on the complements of *get* would appear to involve some semantic notion of physical or metaphorical location (i.e. a further semantic constraint).’ Given the status of (45a), I take it that the reviewer means that, for some reason, examples involving *a*-phrases must involve some notion of location. As far as I can tell, this is correct.

Developing this line of thought further, we should note the following three points. First, one cannot plausibly argue against the probative value of (46) by alleging that we are basically dealing with a prepositional phrase (*from the action*) that is merely modified by *a safe distance*. This is because the prepositional phrase by itself cannot stand in the relevant place:

(47) *The sergeant got the guard from the action.

We also know that *distance* subcategorizes for *from*:

(48) A safe distance from the action would be about ten miles.

Secondly, examples of *a*-phrases taken by *get* can be multiplied indefinitely as long as the constraint about location is observed:

- (49)
- a. The sergeant got the guard a mile from the action
 - b. The sergeant got the guard a long way from the action
 - c. The sergeant got the injured guard a good way towards their base.
 - d. The sergeant got the injured guard a good part of the way.
 - e. The sergeant got the injured guard a good part of the distance.

Thirdly, although a full analysis of the relevant data is beyond the scope of the present article, it is possible that *get* subcategorizes for stage-level predicates in this position. Bearing in mind the rough initial distinction between adjectives and prepositional phrases, on the one hand, and *a*-phrases, on the other, it is notable that, while adjectives and prepositional phrases are frequently stage-level, it has been claimed that predicate nominals, including *a*-phrases, are quite generally individual-level (Carlson 1977; Chierchia 1995). Could it be that phrases like *a safe distance from the action* are stage-level and thus form exceptions to this generalization? They can certainly act as the codas of *there*-expletive sentences, to use a classic diagnostic (Milsark 1977):

- (50)
- a. There are some inexperienced guards a safe distance from the action.
 - b. *There are some inexperienced guards malingerers/observers.

And examples of *got* with individual-level adjectives and prepositional phrases in the relevant position do not sound very good:

- (51)
- a. ??The sergeant got the guard intelligent/brave/tall/American/Catholic/from Italy.
 - b. The sergeant got the guard awake/drunk/drowsy/confused/angry/calm/on the wall.

I will not explore this possibility further here. But it seems promising. Bearing this in mind, it is worth stressing that the distinction between individual-level and stage-level predicates is, of course, a semantic one.

The last verb I will consider in this survey is *wax*. Some history will be relevant here. According to the *Oxford English Dictionary*,²⁰ this verb started out life in Anglo-Saxon times meaning ‘grow’ and developed various other senses, including ‘become’. This latter sense, like most of the senses of this word, is now archaic, but it survived into the 19th Century:

- (52) The howl of contumely...gradually waxed fainter.
(1831, T. B. Macaulay, ‘Moore’s Life Byron’, in his *Essays*)
- (53) Mr. Chuckster waxed wroth at this answer.
(1841, C. Dickens, *Old Curiosity Shop* ii. xxxviii. 6)

These days, usage of the word is almost confined to talking about the moon growing larger, plus one interesting new sense, which dates from the 19th Century:

- (54) To speak or write (increasingly) in the manner specified; esp. in *to wax lyrical*, *to wax eloquent*.

The first attested example of this is from 1842; the sense was added to the dictionary only in 2006.

In this sense, this word is alleged to take only adjective phrases as complements (Dalrymple 2017: 34). Here are some data:

- (55) a. Fred waxed lyrical.
b. Fred waxed eloquent.
c. Fred waxed rhetorical.
d. Fred waxed poetic.
e. Fred waxed nostalgic.
f. Fred waxed enthusiastic.
- (56) a. *Fred waxed in lyrical tones.
b. *Fred waxed in eloquent terms.
c. *Fred waxed in a rhetorical manner.
d. *Fred waxed in poetic language.
e. *Fred waxed with nostalgic longing.
f. *Fred waxed with enthusiastic gestures.

²⁰ *Oxford English Dictionary*, “wax (v.1),” <https://doi.org/10.1093/OED/8402442679>, retrieved 18 November 2025.

- (57) a. *Fred waxed a lyricist.
 b. *Fred waxed an eloquent man.
 c. *Fred waxed a rhetor/orator.
 d. *Fred waxed a poet.
 e. *Fred waxed a nostalgic person.
 f. *Fred waxed an enthusiast.

Even more oddly, *wax* in this sense seems to be restricted not merely to speech but to “the domain of *enthusiastic* speech” (Petré 2012: 33, my emphasis):

- (58) a. wax eloquent
 b. wax enthusiastic
 c. wax poetic
- (59) a. *wax tongue-tied
 b. *wax bored
 c. *wax prosaic

It might be unclear to what extent one can draw broad theoretical conclusions on the basis of such an extremely idiosyncratic lexical item, but the data surveyed so far demand an explanation. In particular, we see here an alleged example of selection on the basis of syntactic category. We will have to see if this stands up.

It is already evident that this is not an example of selection *just* for syntactic category. The data in (58) and (59) make that clear. Could there be semantic factors behind the data in (55)–(57) too?

I suspect there are. Once more, I will not be able to give this matter the attention it deserves within the confines of the current article, but as a basis for further research, I make the following informal suggestion:

- (60) ‘*S* waxed ϕ ’ is grammatical and true iff ‘*S* became ϕ ’ is grammatical and true (where to be ϕ is to be enthusiastic or eloquent and ϕ is a stage-level predicate) and *S*’s being ϕ was manifested in *S*’s speech or writing.

The restriction to stage-level predicates rules out *a*-phrases and other predicate nominals (Carlson 1977; Chierchia 1995), as observed above. (The exceptions to this generalization that we recently noted, like *a safe distance from the action*, are not relevant here, since they do not have anything to do with enthusiastic speech.) This takes care of the data in (57). As for the status of the prepositional phrases in (56), I suggest that the inclusion in the semantics of *wax* of an element like the semantics of *become* takes care of these. The word *become* is evidently very picky about which prepositional phrases it combines with, and it does not seem to combine well with examples pertaining to enthusiastic speech:

- (61) a. *Fred became in lyrical tones.
 b. *Fred became in eloquent terms.
 c. *Fred became in a rhetorical manner.
 d. *Fred became in poetic language.
 e. *Fred became with nostalgic longing.
 f. *Fred became with enthusiastic gestures.

This may be sufficient to deal with the examples in (56), and similar. A note in the *Oxford English Dictionary* says that the sense of *wax* that we are discussing likely developed, diachronically, from the sense ‘become’ that we saw in (53).²¹

Certain modifiers like *very* have also been traditionally taken to subcategorize for a subclass of the three kinds of predicates under discussion in this subsection. The word *very*, for example, seems at first sight to modify adjectives but not prepositional phrases (Maling 1983: 261):

- (62) a. The birds were very noisy.
 b. *The birds were very in the tree.

However, the empirical picture here seems to be rather muddy in a way that is now familiar, since some examples of this word modifying prepositional phrases are fine (Maling 1983: 261). The following naturally occurring examples were consulted online on 19 November 2025:

- (63) This episode is very off the wall.²²
 (64) Yeah, she did her thesis on postmodernism, but she’s very at home in Victorian lit too—she’s actually quite the Tennyson buff.²³
 (65) When I started, I wasn’t very in shape. I couldn’t even do a burpee and could barely run.²⁴
 (66) Dancers are very in tune with their emotions.²⁵

But *very* (as opposed to *very much*) does not, as far as I know, modify *a*-phrases:

- (67) *This episode is very an anomaly.
 (68) *She’s very a connoisseur of Victorian literature.
 (69) *I wasn’t very an athlete.

²¹ *Oxford English Dictionary*, “wax (v.1),” <https://doi.org/10.1093/OED/8402442679>, retrieved 18 November 2025.

²² <https://podcasts.apple.com/us/podcast/episode-13-your-onomato-mother/id452488806>.

²³ <https://idioms.thefreedictionary.com/be+at+home>.

²⁴ <https://www.anchorfitness.training/testimonials>.

²⁵ <https://www.merriam-webster.com/sentences/intune>.

(70) *This dancer is very a perceptive person.

An anonymous reviewer for this journal (building on remarks in Maling 1983 and other literature) suggests that the reason for this restriction might be that *very* subcategorizes for gradable predicates. In fact, given its meaning, it seems that *very* could scarcely do anything but subcategorize for gradable predicates. Now we cannot say that *a*-phrases are not gradable at all (which would be one easy move here), since we have examples like the following:

(71) This episode is very much an anomaly.

(72) She is quite a connoisseur of Victorian literature.

But it is likely that there are different ways of spelling out gradability in semantic terms, with the way it is done in adjectives differing from the way it is done in nouns and verbs (Morzycki 2012; 2016; Rett 2013), for example. All we need to explain the inability of *very* to combine with *a*-phrases is the plausible hypothesis that the expression of gradability in *a*-phrases too differs in semantic terms from the expression of gradability in adjectives (and in certain prepositional phrases, apparently): *very* might be targetting something (degrees, for example) that does not exist in *a*-phrases (or which is deployed differently in these phrases), thus making it semantically incompatible with the latter.

We can conclude, then, that casting a fresh eye on the facts concerning English verbs and modifiers makes it seem unlikely that there is a problem here for the proposal that subcategorization is carried out on the basis of semantic type combined with further semantic properties. This whole subject is in need of a thorough review.

4 The basic coordination data

4.1 Predicative position

This subsection will deal with the predicate coordination data most often encountered in the literature, which is constituted by examples in which the conjoined predicates are in predicative position. We are dealing with examples like the following, whose conjoined components were discussed at length in the previous section:

(73) John is disreputable and of no fixed abode. (“AP and PP”)

(74) John is disreputable and a vagabond. (“AP and *a*-phrase”)

(75) Nietzsche is a philosopher and beyond good and evil. (“*a*-phrase and PP”)

One sometimes also encounters examples like the following, with the claim that they involve a VP being coordinated with another category (Sag et al. 1985: 118):

(76) I am hoping to get an invitation and optimistic about my chances.

However, I agree with Bruening & Al Khalaf (2020: 7), who label participles in constructions like this APs. I assume that, whatever the internal syntactic and semantic complexity of participles, they are basically verbal adjectives when it comes to outward-facing semantics. I will not distinguish them further from other adjectives in this article.²⁶

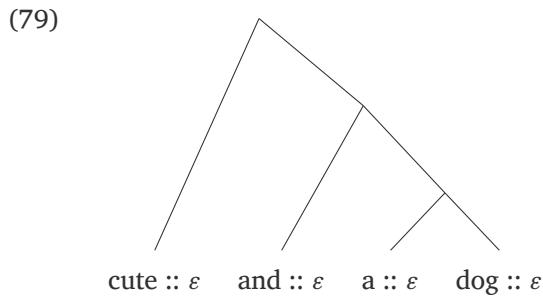
So how should these data be analysed?²⁷ It will come as no surprise by now to learn that I posit a sense of the word *and* that takes two arguments of type $\langle \text{eit}, \text{eit} \rangle$ and maps them to another constituent of the same type. Unlike verbs like *get*, it does not impose any requirements on its arguments beyond their overall type:

(77) $\text{/and/} :: \lambda F_{\langle \text{eit}, \text{eit} \rangle} . \lambda G_{\langle \text{eit}, \text{eit} \rangle} . \lambda f_{\langle \text{e}, \text{it} \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ F(\lambda y . \lambda t'' . \top)(x)(t) \ \& \ G(\lambda y . \lambda t'' . \top)(x)(t)$
 $:: E_R \ E_L$

Suppose we are dealing with (78):

(78) Fido is cute and a dog.

The version of *and* in (77) will combine with *cute* and *a dog* to create the following structure.²⁸



A straightforward calculation produces the following semantic value for this structure:

(80) $\lambda f_{\langle \text{e}, \text{it} \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ \text{cute}(x)(t) \ \& \ \text{dog}(x)(t)$

²⁶ Once more, I would like to stress that I am using terms like *adjective* and *participle* in an informal way, for convenience. Such category labels, of course, have no official status in the current theory.

²⁷ It has been adequately shown, I think, in line with other arguments concerning coordination in general, that it will not help us to suppose that there is ellipsis here and that what appear to be coordinations of two predicates really involve discontinuous constituents (Peterson 2004: 648; Bruening & Al Khalaf 2020: 9). I will review other previous solutions in section 5.

²⁸ I assume a binary-branching structure for coordination, as defended by Zhang (2010) and others. But see Neeleman et al. (2023) for a different view.

This is of type $\langle \text{eit}, \text{eit} \rangle$, meaning that (78) will now work just like (16) (*Fido was cute*). Other cases like those in (73)–(75) will work analogously.

The denotation in (77) is of type $\langle \text{eiteit}, \langle \text{eiteit}, \text{eiteit} \rangle \rangle$. This is a unique type in the current system. Arguably any theory would need a version of *and* that maps two predicate meanings to a third predicate meaning in order to deal with these data. But the current theory allows us to cash out the hypothesis that the data make obvious and intuitive: that coordination in these cases is based on identity of semantic type.

4.2 Attributive position

A tricky problem (for all frameworks) arises in connection with conjunctions of predicates in attributive position. Sag et al. (1985: 141) note that while (81) is fine, (82) is unacceptable:

(81) Terry turned out to be longwinded and a bully.

(82) *The longwinded and a bully man was my brother.

But (83), of course, is fine:

(83) The longwinded and boring man was my brother.

More generally, a conjunction of adjectives in attributive position is fine, but other combinations of the predicates we have been discussing are ungrammatical.

The generalization about what conjunctions are possible in attributive position might also be expressed as follows: all and only predicates that can appear by themselves in attributive position can appear as conjuncts in attributive position.²⁹ Just as the (a) examples in (84) and (85) are ungrammatical, so the corresponding (b) examples are ungrammatical; just as (86a) is grammatical, so is (86b) grammatical.

(84) a. *the a bully man
b. *the longwinded and a bully man

(85) a. *the in danger man
b. *the longwinded and in danger man

(86) a. the boring man
b. the longwinded and boring man

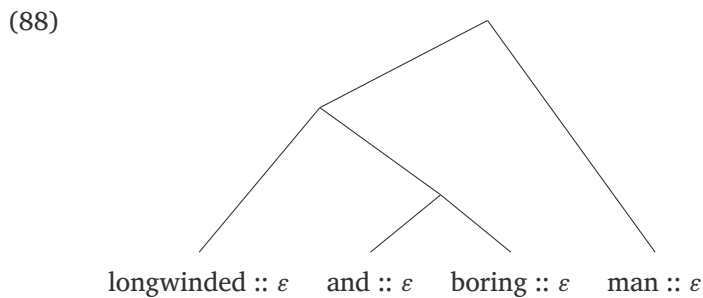
²⁹ Actually there is a small class of apparent exceptions to this generalization, of which *the once and future king* is the most prominent example (Bruening & Al Khalaf 2020; Patejuk & Przepiórkowski 2023; Bruening 2025). I do not at present have anything detailed to say about these cases, although I make a brief speculation in Section 6. I will assume that the generalization expressed in the main text is near enough accurate to be worth pursuing.

This suggests that the ability to combine with a syntactic argument to their right might be the decisive factor in allowing adjectives to appear in conjunctions in attributive position.³⁰

As mentioned in sections 2.1.1 and 2.1.3, I propose that *and* and *or* can subcategorize for constituents bearing particular selector features. In the case of (86b), for example, we will be dealing with the following lexical entry for *and*:

$$(87) \quad \text{/and/} :: \lambda F_{\langle \text{eit}, \text{eit} \rangle} . \lambda G_{\langle \text{eit}, \text{eit} \rangle} . \lambda f_{\langle \text{e}, \text{it} \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ F(\lambda y . \lambda t'' . \top)(x)(t) \ \& \\ G(\lambda y . \lambda t'' . \top)(x)(t) \\ :: E_{R, E_R} \ E_{L, E_R} \ E_R$$

This is identical to the lexical entry in (77) except for the selector features. I will shortly make a few remarks about how to unify the two and prevent unwelcome variants. But first note that, given the selector features and the meaning, this version of *and* can only assemble adjectives as conjuncts: adjectives are the only items of type $\langle \text{eit}, \text{eit} \rangle$ (this type being required due to the $\lambda F_{\langle \text{eit}, \text{eit} \rangle} . \lambda G_{\langle \text{eit}, \text{eit} \rangle} \dots$) that can bear E_R features. When External Merge merges in an adjective on the basis of this kind of selector feature, the E_R feature on the adjective that was selected for will be deleted as part of the Merge operation, as mentioned in section 2.1.3. The last selector feature, E_R , will merge in a noun for the whole conjunction to modify. This means that, given adjective and noun denotations like those in (7), this version of *and* allows us to build the following tree:



A straightforward calculation produces the following denotation for this example:

$$(89) \quad \lambda x . \lambda t . \text{man}(x)(t) \ \& \ \text{boring}(x)(t) \ \& \ \text{longwinded}(x)(t)$$

³⁰ An anonymous reviewer points out that a small number of prepositional phrases *can* actually appear in conjunctions in attributive position: *an aggressive and in-your-face challenge*, *an unexpected and off-the-wall proposal*. But these exceptions actually support the overall line being taken here, since, again exceptionally, they can appear outside conjunction with what looks like a syntactic argument to their right: *an in-your-face challenge*, *an off-the-wall proposal*. I suspect that these cases, which are highly idiomatic (i.e. non-compositional), are listed as such, as idiom chunks, in the lexicon, complete with exceptional E_R features that enable them to behave as they do.

This is of type $\langle e, it \rangle$, as required; the constituent in (88) could now be taken as argument by a or (in a larger fragment) by a quantifier.

We now have two versions of *and*, in (77) and (87). The meaning was identical in both cases. The only difference was in the selector features, which were as follows:

$$(90) \quad E_R \ E_L$$

$$(91) \quad E_{R,E_R} \ E_{L,E_R} \ E_R$$

The selector features in (90) assemble the conjuncts and do nothing else; they leave the conjunction free of uninterpretable features and thus able to be taken as an argument by the copula or another linking verb. Note that a conjunction constructed by means of this set of features could not appear in attributive position, since neither it nor any noun has any selector features to bring about the necessary Merge. On the other hand, the features in (91), as we have just seen, assemble the conjunction and also merge in a noun to its right that it modifies. The guiding intuition in the second case is that a conjunction of adjectives in attributive position (and we have made sure that we will only get adjectives here) will behave just like an individual adjective in that position: it will have an E_R feature to merge in the noun to be modified.

This is all very well and might even lay claim to a certain naturalness. But we must rule out certain other conceivable sequences of selector features. What would happen, for example, if *and* with the current meaning could have the following selector features?

$$(92) \quad E_R \ E_L \ E_R$$

This would allow phrases like (84b) (**the longwinded and a bully man*), since the requirement for E_R features on the first two constituents to be merged has now been dropped.

I suggest that the permissible selector features on the current version of *and* be indicated as in the following lexical entry:

$$(93) \quad \text{/and/} :: \lambda F_{\langle eit, eit \rangle} . \lambda G_{\langle eit, eit \rangle} . \lambda f_{\langle e, it \rangle} . \lambda x . \lambda t . f(x)(t) \ \& \ F(\lambda y . \lambda t'' . \top)(x)(t) \ \& \\ G(\lambda y . \lambda t'' . \top)(x)(t) \\ :: E_{R,\gamma} \ E_{L,\gamma} \ \gamma$$

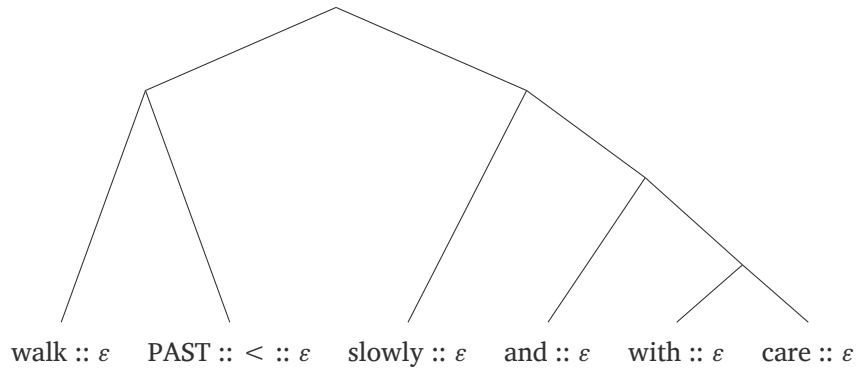
Here the symbol γ is a variable over the three possibilities that obtain when it comes to selector features: E_R , E_L , and the empty sequence ε . That gives us three permissible options: (90), with ε put in for γ ; (91), with E_R ; and (94), with E_L .

$$(94) \quad E_{R,E_L} \ E_{L,E_L} \ E_L$$

This last will enable conjunctions of prepositional phrases (and, in a larger fragment, restrictive relative clauses) after nouns, as in the following examples:

features; and the complex adverbial phrase will be placed to the right of the verb. With these options selected, the following tree is easily built:

(101)



A straightforward calculation produces the following denotation for (101):

(102) $\lambda e.\lambda t.\text{walking}(e) \ \& \ < (\text{CUL}(e))(t) \ \& \ \text{with}_2(e)(c) \ \& \ \text{slow}(e)$

This is of type $\langle s, it \rangle$ (just like *walked*), ready to be taken as an argument by little *v*, which will in turn merge in *John*, as in (27).

I have marked down the final selector feature of this version of *and* as “ E_L/E_R ” since it seems that, in appropriate circumstances, a conjunction of type $\langle \text{sit}, \text{sit} \rangle$ items can go to the left of a verb as well as to the right:

(103) John slowly and carefully walked across the minefield.

(104) ?John slowly and with care walked across the minefield.

There seems to be no need to have the selector features on this version of *and* target items with particular selector features, as was done in the last subsection for *and* in attributive position, because we are not here faced with a similar problem of something being of the right type to appear in a particular position but not being able to. The issue will ultimately depend on the status of examples like (104), which tend to be slightly worse than examples like (103). To me, at least, (104) seems slightly awkward but basically fine. If a consensus emerged that examples like this are ungrammatical, then something more complex, along the lines of the solution in section 4.2, could be developed in this case too.

There is one more point about the data in the adverbial realm that is worth making. Patejuk & Przepiórkowski (2023) and Bruening (2025) cite the following sentence as an example of an adverb (*worse*) coordinated with a DP:

(105) Many in DC behave [this way or worse].

From the current perspective, nothing actually has the syntactic category DP, of course. All that is necessary is that *this way* should be of the same semantic type as *worse*. In the current version of the theory, that will be $\langle \text{sit}, \text{sit} \rangle$. I suspect that the way forward here is to say that *this way* is just recorded in the lexicon as an idiom chunk with an appropriate meaning of this type. (The same apparently holds true of *that way*.) This approach is made plausible, I think, by the fact that the kind of meaning with which we are dealing cannot be computed compositionally in the case of variants in which near synonyms are switched in for *way*:

- (106) a. Many in DC behave in this manner.
 b. *Many in DC behave this manner.
- (107) a. Many in DC behave in this fashion.
 b. *Many in DC behave this fashion.

Traditional approaches face a tricky question about syntactic categories here: roughly, “What theory of coordination of syntactic categories will allow AdvP to be coordinated with DP without overgenerating?” But this question dissolves instantly when the current approach is adopted.

5 Previous accounts

In this section I will make some remarks, necessarily brief, on two prominent previous accounts of the data that I deal with in this article. But first, in order to provide a benchmark, I will describe what I take to be the main virtue of the approach advocated in the current article. That virtue is economy. My basic contention is that the versions of *and* that are in play here select for their arguments on the basis of semantic type, not on the basis of syntactic categories, which play no role in the current system. This factor by itself allows for an explanation of the data concerning conjunctions in predicative position (section 4.1) and adverbial position (section 4.3). The data concerning conjunctions in attributive position (section 4.2) were more complex, but even those were dealt with by means of a simple expedient that was based on theoretical constructs already posited for different reasons: the idea was that, as well as selecting for semantic type, *and* could also select for particular kinds of selector features. It should be emphasized once more that the selector features used in the current system are direct descendants of features used in previous versions of Minimalist syntax—see section 2.1.1 for details. And semantic types are inherent properties of the objects and functions that contemporary semanticists take to be the denotations of linguistic items in the mainstream tradition of compositional semantics that goes back through Lewis (1970) and Montague (1970; 1973) to Frege (1891). The idea is, then, that the data under consideration either fall out directly from the very nature of the current system (in the case of selection for semantic type) or are accounted for by the judicious use of elements that are already posited in the current system (and in previous versions of Minimalist grammar) for other reasons.

This contrasts with a prominent group of previous accounts that add an extra syntactic category (or syntactic feature) specially to deal with the current data. Goodall (1987: 44), for example, proposes that we account for the data at hand by postulating not only normal syntactic categories but also “superordinate archicategories”, the idea being that coordination can coordinate either. In the predicative cases, for example, there is an archicategory Predicate and in the adverbial cases there is an archicategory Manner: adjectives, *a*-phrases, and certain PPs belong to Predicate, while adverbs and other PPs belong to Manner. Similar accounts have been given by Bowers (1993), who posits a category PrP for predicates, and Bruening & Al Khalaf (2020), who posit “supercategories” Pred and Mod. Sag et al. (1985) posit a feature PRD to be shared by NP, AP, and PP, which is selected for in predicative coordination cases; a similar feature MANNER is used in the adverbial cases. I submit that all these approaches compare unfavourably with mine in terms of economy, since they make up ad hoc categories or features to deal with the coordination data rather than using resources already present in their systems for independent reasons. It is also the case, I would contend, that making up a new category or a new feature to account for these cases is in a sense too easy. Patejuk & Przepiórkowski (2023: 331) maintain that this approach is unfalsifiable: given any odd-looking coordination, a new feature or category could be made up to accommodate it, with no obvious way of constraining the theory.³²

The second group of solutions on which I will comment here is that which uses grammatical functions (Dik 1968; Peterson 1981; 2004). The idea here is that coordination is capable of targetting two constituents whose grammatical functions are the same. Peterson (2004: 650), for example, building on the tradition of Lexical Functional Grammar (LFG), makes use of a grammatical function XCOMP that is shared by all the items that are capable of being coordinated in predicative position.

However, theories based on grammatical functions face a problem in this context from the phenomenon of heterofunctional coordination, as noted (for example) by Zhang (2007) and Przepiórkowski et al. (2025). Heterofunctional coordination, as the name implies, is when conjuncts would be assigned different grammatical functions by theories that trade in these things. Relevant data were first introduced into the literature by Browne (1972). Here are some examples, taken from Przepiórkowski (2022: 276, 280; Polish) and Citko & Gračanin-Yuksek (2013: 18; Romanian), respectively:

(108) Tu krytykują [wszyscy i wszystkich].

here criticize.PL all.NOM.PL and all.ACC.PL

‘Here everyone criticizes someone or other and everyone is criticized by someone or other.’

³² For further critical discussion of this approach, see Peterson (1981: 452). One could, of course, attempt to constrain the theory by means of shared semantics, which is presumably implicit in the names chosen for the categories and features just discussed. But then that becomes a version of the current theory, more or less economical according as syntactic features are maintained in the overall system.

- (109) [Cine și cu cine] s-a întâlnit?
 who and PREP who REFL-AUX met
 ‘Who met whom?’

In these examples we see subjects coordinated with objects. It is hard to see how the theory that says that coordination depends on identity of grammatical function could deal with these data. The only possibility that springs to mind is that it could maintain that all examples of this kind are in fact elliptical structures in which two whole clauses are conjoined, with ellipsis creating the illusion of the conjunction of smaller constituents like those bracketed in (108) and (109). This is a strategy that has in fact been used successfully to account for some superficially similar examples in English (Przepiórkowski et al. 2025). But a substantial literature has grown up around this question and has argued, for several languages and with the aid of a variety of diagnostics, that we are at least sometimes dealing with conjunctions of smaller constituents like those in the above examples. This has been argued for cases in the following languages: Bulgarian, Romanian, Polish, Russian, and Croatian (Lipták 2011; Citko & Gračanin-Yuksek 2013); Hungarian (Lipták 2011); Chinese (Zhang 2007); and German (Przepiórkowski et al. 2025).³³ It is likely, then, that theories that rely on grammatical functions face problems of this kind that do not affect the current theory.

6 Conclusion

The main advantage of the current analysis of the data under consideration, that of economy, has been stressed in the previous section.

One additional advantage of the current analysis is that it completely avoids the question of what the syntactic category of conjunctions should be, which is a problem that has dogged work on conjunction at least since Sag et al. (1985: 141). In the current framework, of course, nothing has any syntactic category.

As well as the two cases on which I have concentrated in this article, there are several other kinds of example that have been claimed to present problems for the thesis that coordination is restricted to like syntactic categories. I will not have the space to discuss them in detail here and will have to leave their detailed analysis to future research. But it is as well to acknowledge them. Here is a list (not necessarily complete) with some brief comments, many of which are speculative.

- i. I have already dealt with (105), repeated here:

- (110) Many in DC behave [this way or worse].

³³ The literature on this topic is now quite large. I have given only a few representative citations. For a convenient introduction to the subject and a more complete bibliography, see the first section of Przepiórkowski et al. (2025).

As I said, I suspect that the way forward in this case is simply to list *this way* and *that way* in the lexicon as idiom chunks with appropriate adverbial meanings. So the two conjuncts in cases like (110) would have the same semantic types.³⁴

- ii. There are examples like the following, which apparently involve arguments conjoined with modifiers (Grosu 1985):

(111) John eats only pork and only at home.

But this kind of example is widely assumed to be basically a conjunction of whole clauses, with ellipsis giving an illusion of smaller conjuncts (Bruening & Al Khalaf 2020; Patejuk & Przepiórkowski 2023). In this kind of case, the two actual conjuncts will have the same semantic types, then.

- iii. Subordinate clauses are sometimes conjoined with DPs after verbs (Dik 1968: 28):

(112) I want to emphasize this point and also that you should never forget what your father told you.

A puzzling variant of this kind of case is that in which one of the conjuncts would be ungrammatical in the relevant position by itself (Sag et al. 1985):

- (113) a. You can depend on my assistant and that he will be on time.
 b. *You can depend on that he will be on time.

An obvious and potentially fruitful approach to (112) and (113a) would be to make the clausal conjunct equivalent in meaning to a nominal like *the fact that...*, perhaps with the aid of unpronounced structure. This is similar to the approach of Bruening & Al Khalaf (2020). But there is a difficult problem here about why (113b) is not thereby rendered grammatical. See Patejuk & Przepiórkowski (2023) and Bruening (2025) for discussion. In my opinion, the most promising line to take here is to say that (113a) is not really grammatical but merely benefits from an illusion of grammaticality brought on by processing factors: Kim & Lu (2024) plausibly argue that, to the extent that (113a) seems grammatical, this is due to our memory of the exact syntactic environment of the second conjunct having faded by the time we process it, a phenomenon known in other cases and referred to as *structural forgetting* (Gibson & Thomas 1999). If this is correct, we are left only with examples like (112). Making the clausal conjuncts in these examples equivalent in meaning to nominals would make the two conjuncts be of the same semantic type.

³⁴ It must be admitted, however, that there are other examples of apparent coordination of DPs with adverbs that look like they constitute compositionally interpreted structures. An example is *Immunity may last ten years or longer* (Patejuk & Przepiórkowski 2023: 331), where of course we could have any number in place of *ten*. I will have to leave these cases for future research.

- iv. An entire class of counterexamples is constituted by the title of a classic fantasy novel by T. H. White (and trivial variants; Bruening & Al Khalaf 2020):

(114) *The Once and Future King*

This looks like a conjunction of an adverb and an adjective. At first sight, this seems to be another example in which one of the conjuncts is not grammatical in the relevant environment when it is not in a conjunction:

(115) ??the once king

However, Patejuk & Przepiórkowski (2023) and Przepiórkowski & Patejuk (2024) have cited numerous naturally occurring examples in which phrases like this are in fact perfectly fine:

(116) (*Talking about Charles III, shortly after the death of Elizabeth II...*) At last year's global climate talks in Glasgow, the now monarch said: "We know what we must do."

This opens up the possibility, considered sympathetically by Przepiórkowski & Patejuk (2024), that words like *once* and *now* might in fact have both adverbial and adjectival uses (with the availability of the latter being subject to certain specific conditions, presumably, given that (115) does not sound very good in isolation). In the terms of the current article, this would be expressed by saying that words like *once* and *now*, as well as their adverbial senses, also have meanings that are of type $\langle \text{eit}, \text{eit} \rangle$. So once more we would have conjuncts that were of the same semantic type.

- v. Patejuk & Przepiórkowski (2023: 344) point out that verbs that combine with DPs and PPs separately can also combine with mixed conjunctions of these categories:

(117) He then stated a number of ways people can fight [[the intolerance] and [against those who twist religion to use it for evil]].

It is possible that these are to be explained by appealing to the idea that direct objects and other internal arguments are added to verbs by means of functional heads analogous to little *v* (Borer 2005; Pietroski 2007; Alexiadou 2014; Lohndal 2014; Champollion 2015). Work in this tradition assumes that these combinations of functional head plus direct object take verbs as arguments. At least some PPs could be made to do the same. Some work would have to be done to prevent overgeneration. But if this worked out, it would mean that the conjuncts in (117) were of the same semantic type.

- vi. The same thing happens with verbs that take CPs and PPs separately (Patejuk & Przepiórkowski 2023: 344):

(118) We all believe [[in positive energy] and [that what you give comes back]].

The examples given by Patejuk & Przepiórkowski all involve propositional attitude verbs. It is likely that *in* + DP in this configuration denotes a proposition. Very roughly, *in* would take a DP that denoted such-and-such a thing and give back the proposition that such-and-such a thing exists. (Alternatively, there might be hidden structure.) This might seem exotic at first sight, but in fact it is already widely assumed that intensional transitive verbs (as in *I want a sloop*) take propositions as internal arguments, in spite of the unpromising surface syntax (Quine 1956; den Dikken et al. 1996; 2018; Parsons 1997). This would just be one more example. So it is likely that the conjuncts here are of the same semantic type.

- vii. And likewise with verbs that take infinitive phrases and DPs or infinitive phrases and CPs headed by *that* (Patejuk & Przepiórkowski 2023: 344):

(119) We teach them [manners and [to be respectful]].

(120) You have taught me [[to rest physically] and [that it is okay to work hard]].

Since infinitive clauses have traditionally been analysed as including subjects, sometimes silent ones, it is likely that the two conjuncts in (120) are propositional. As for (119), it is notable that all the examples of this kind given by Patejuk & Przepiórkowski involve either *teach* or *want*. The latter is a paradigm case of an intensional transitive verb. It is likely that *teach* is intensional too: *I taught him the positions of Phosphorus* does not imply that I taught my student how to locate Venus in the evening. It is likely, then, that the conjuncts in this kind of example are all propositional. This means that they would be of the same semantic type.

As far as I know, no-one has given detailed syntactic and semantic analyses of all these cases. I hope that these remarks make it seem plausible that the kind of analysis spelled out in this article for predicates and modifiers will be able to be extended to the above cases after further research.

Of course, theories that use autonomous syntactic categories could in principle also maintain that coordination relies on semantic type; but then they would be in the awkward position of saying that syntactic combination relies sometimes on syntactic category and sometimes on semantic type, with no evident principled mechanism for determining which option applies.

By presenting an economical analysis that in important respects was simply a direct consequence of the overall framework, this paper has arguably shown an advantage of adopting that framework, the framework of Categoriless Meaning-Dependent Grammar.

Abbreviations

ACC	accusative
AUX	auxiliary
NOM	nominative
PL	plural
PREP	preposition
REFL	reflexive

Funding information

This work was funded by a Leverhulme Major Research Fellowship (MRF-2023-126).

Acknowledgements

I am extremely grateful to three reviewers for *Glossa*, one of whom outed himself as Adam Przepiórkowski. Their detailed and insightful comments have substantially improved this article. Thanks also to my editor at *Glossa*, Agata Renans, the production editor, Johan Rooryck, and the production staff at SiliconChips.

This work was funded by a Leverhulme Major Research Fellowship (MRF-2023-126). I am extremely grateful to the Leverhulme Trust for their generous support.

I have given versions of this paper at the ‘Philosophy of Language and Linguistics’ conference at the Inter-University Centre in Dubrovnik in September 2025 and at the ‘Role of Semantics in Syntax’ conference at Magdalen College, Oxford, in June 2026. I am grateful to the audiences on both occasions and especially, for their comments and questions, to Víctor Acedo-Matellán, Chris Barker, Mark Bowker, Tamara Dobler, Ricardo Etxepare, Michael Glanzberg, Kerstin Hoge, Vera Hohaus, Stefan Kaufmann, Gillian Ramchand, Craige Roberts, Uli Sauerland, Barry Smith, and Danfeng Wu.

Competing interests

The author has no competing interests to declare.

References

- Adger, David. 2003. *Core syntax*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780199243709.001.0001>
- Alexiadou, Artemis. 2014. Roots don’t take complements. *Theoretical Linguistics* 40. 287–297. DOI: <https://doi.org/10.1515/tl-2014-0012>
- Bar-Hillel, Yehoshua. 1953. A quasi-arithmetical notation for syntactic description. *Language* 29(1). 47–58. DOI: <https://doi.org/10.2307/410452>

- Borer, Hagit. 2005. *Structuring sense: Volume II: The normal course of events*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/acprof:oso/9780199263905.001.0001>
- Bowers, John. 1993. The syntax of predication. *Linguistic Inquiry* 24(4). 591–657.
- Browne, E. Wayles. 1972. Conjoined question words and a limitation of English surface structures. *Linguistic Inquiry* 3(2). 223–226.
- Bruening, Benjamin. 2025. Selectional violations in coordination (a response to Patejuk and Przepiórkowski 2023). *Linguistic Inquiry* 56(3). 439–483. DOI: https://doi.org/10.1162/ling_a_00506
- Bruening, Benjamin & Al Khalaf, Eman. 2020. Category mismatches in coordination revisited. *Linguistic Inquiry* 51(1). 1–36. DOI: https://doi.org/10.1162/ling_a_00336
- Carlson, Gregory Norman. 1977. *Reference to kinds in English*. Amherst: University of Massachusetts dissertation.
- Champollion, Lucas. 2015. The interaction of compositional semantics and event semantics. *Linguistics and Philosophy* 38. 31–66. DOI: <https://doi.org/10.1007/s10988-014-9162-8>
- Chierchia, Gennaro. 1995. Individual-level predicates as inherent generics. In Carlson, Gregory & Pelletier, Francis (eds.), *The generic book*, 176–223. Chicago: University of Chicago Press.
- Chomsky, Noam. 1957. *Syntactic structures*. The Hague: Mouton. DOI: <https://doi.org/10.1515/9783112316009>
- Chomsky, Noam. 2008. On phases. In Freidin, Robert & Otero, Carlos P. & Zubizarreta, Maria Luisa (eds.), *Foundational issues in linguistic theory*, 133–166. Cambridge, MA: MIT Press. DOI: <https://doi.org/10.7551/mitpress/7713.003.0009>
- Chomsky, Noam. 2013. Problems of projection. *Lingua* 130. 33–49. DOI: <https://doi.org/10.1016/j.lingua.2012.12.003>
- Citko, Barbara & Gračanin-Yuksek, Martina. 2013. Towards a new typology of coordinated wh-questions. *Journal of Linguistics* 49. 1–32. DOI: <https://doi.org/10.1017/S0022226712000175>
- Clark, Romane. 1970. Concerning the logic of predicate modifiers. *Noûs* 4(4). 311–335. DOI: <https://doi.org/10.2307/2214677>
- Collins, Chris & Stabler, Edward. 2016. A formalization of minimalist syntax. *Syntax* 19(1). 43–78. DOI: <https://doi.org/10.1111/synt.12117>
- Cresswell, M. J. 1996. *Semantic indexicality*. Dordrecht: Kluwer. DOI: <https://doi.org/10.1007/978-94-015-8696-2>
- Dalrymple, Mary. 2017. Unlike phrase structure category coordination. *Bergen Language and Linguistics Studies* 8. 33–55. DOI: <https://doi.org/10.15845/bells.v8i1.1332>
- den Dikken, Marcel. 2006. *Relators and linkers: The syntax of predication, predicate inversion, and copulas*. Cambridge, MA: MIT Press. DOI: <https://doi.org/10.7551/mitpress/5873.001.0001>
- den Dikken, Marcel & Larson, Richard & Ludlow, Peter. 1996. Intensional ‘transitive’ verbs and concealed complement clauses. *Rivista di Linguistica* 8. 331–348.

- den Dikken, Marcel & Larson, Richard & Ludlow, Peter. 2018. Intensional transitive verbs and abstract clausal complementation. In Grzankowski, Alex & Montague, Michelle (eds.), *Non-propositional intentionality*, 46–94. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780198732570.003.0003>
- Dik, Simon C. 1968. *Coordination: Its implications for the theory of general linguistics*. Amsterdam: North-Holland.
- Elbourne, Paul. 2016. Incomplete descriptions and indistinguishable participants. *Natural Language Semantics* 24(1). 1–43. DOI: <https://doi.org/10.1007/s11050-015-9118-8>
- Elbourne, Paul. 2024. A program for eliminating syntactic categories. *Linguistic Inquiry*. Online First. DOI: https://doi.org/10.1162/ling_a_00540
- Elbourne, Paul. 2026. Adjectives without syntactic categories. *Natural Language & Linguistic Theory* 44(17). 1–45. DOI: <https://doi.org/10.1007/s11049-026-09703-w>
- Folli, Raffaella & Harley, Heidi. 2007. Causation, obligation, and argument structure: On the nature of little v. *Linguistic Inquiry* 38(2). 197–238. DOI: <https://doi.org/10.1162/ling.2007.38.2.197>
- Frege, Gottlob. 1891. *Function und Begriff: Vortrag, gehalten in der Sitzung vom 9. Januar 1891 der Jenaischen Gesellschaft für Medizin und Naturwissenschaft*. Jena: Hermann Pohle.
- Gazdar, Gerald. 1980. A cross-categorical semantics for coordination. *Linguistics and Philosophy* 3. 407–409. DOI: <https://doi.org/10.1007/BF00401693>
- Gibson, Edward & Thomas, James. 1999. Memory limitations and structural forgetting: The perception of complex ungrammatical sentences as grammatical. *Language and Cognitive Processes* 14. 225–248. DOI: <https://doi.org/10.1080/016909699386293>
- Goodall, Grant. 1987. *Parallel structures in syntax: Coordination, causatives, and restructuring*. Cambridge: Cambridge University Press.
- Grosu, Alexander. 1985. Subcategorization and parallelism. *Theoretical Linguistics* 12. 231–240. DOI: <https://doi.org/10.1515/thli.1985.12.2-3.231>
- Heim, Irene & Kratzer, Angelika. 1998. *Semantics in generative grammar*. Oxford: Blackwell.
- Kennedy, Christopher. 2007. Vagueness and grammar: The semantics of relative and absolute gradable adjectives. *Linguistics and Philosophy* 30(1). 1–45. DOI: <https://doi.org/10.1007/s10988-006-9008-0>
- Kim, Nayoun & Lu, Jiayi. 2024. Coordination of unlike categories creates grammaticality illusion. *Korean Journal of English Language and Linguistics* 24. 52–61. DOI: <https://doi.org/10.15738/kjell.24..202401.52>
- Kirkpatrick, James & Lewerentz, Lukas. 2025. Objectual quantifier theory. *Journal of Philosophical Logic* 54. 793–855. DOI: <https://doi.org/10.1007/s10992-025-09803-9>
- Lewis, David. 1970. General semantics. *Synthese* 22. 18–67. DOI: <https://doi.org/10.1007/BF00413598>
- Lipták, Anikó. 2011. Strategies of wh-coordination. *Linguistic Variation* 11(2). 149–188. DOI: <https://doi.org/10.1075/lv.11.2.02lip>

- Lohndal, Terje. 2014. *Phrase structure and argument structure: A case study of the syntax-semantics interface*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/acprof:oso/9780199677115.001.0001>
- Maienborn, Claudia. 2005. On the limits of the Davidsonian approach: The case of copula sentences. *Theoretical Linguistics* 31. 275–316. DOI: <https://doi.org/10.1515/thli.2005.31.3.275>
- Maling, Joan. 1983. Transitive adjectives: A case of categorial reanalysis. In Heny, Frank & Richards, Barry (eds.), *Linguistic categories: Auxiliaries and related puzzles*, 253–289. Dordrecht: Reidel. DOI: https://doi.org/10.1007/978-94-009-6989-6_8
- Milsark, Gary. 1977. Toward an explanation of certain peculiarities of the existential construction in English. *Linguistic Analysis* 3. 1–29.
- Montague, Richard. 1970. English as a formal language. In Visentini, Bruno (ed.), *Linguaggi nella società e nella tecnica*, 189–223. Milan: Edizioni di Comunità.
- Montague, Richard. 1973. The proper treatment of quantification in ordinary English. In Hintikka, Jaakko & Moravcsik, Julius & Suppes, Patrick (eds.), *Approaches to natural language: Proceedings of the 1970 Stanford workshop on grammar and semantics*, 221–242. Dordrecht: Reidel.
- Morzycki, Marcin. 2012. The several faces of adnominal degree modification. In Choi, Jaehoon & Hogue, E. Alan & Punske, Jeffrey & Tat, Deniz & Schertz, Jessamyn & Trueman, Alex (eds.), *Proceedings of the 29th West Coast Conference on Formal Linguistics*, 187–195. Somerville, MA: Cascadilla Press.
- Morzycki, Marcin. 2016. *Modification*. Cambridge: Cambridge University Press.
- Neeleman, Ad & Philip, Joy & Tanaka, Misako & van de Koot, Hans. 2023. Subordination and binary branching. *Syntax* 26. 41–84. DOI: <https://doi.org/10.1111/synt.12244>
- Parsons, Terence. 1970. Some problems concerning the logic of grammatical modifiers. *Synthese* 21. 320–324.
- Parsons, Terence. 1997. Meaning sensitivity and grammatical structure. In Chiara, Maria Luisa Dalla & Doets, Kees & Mundici, Daniele & van Benthem, Johan (eds.), *Structures and norms in science*, 369–383. Dordrecht: Kluwer.
- Partee, Barbara. 1989. Binding implicit variables in quantified contexts. In Wiltshire, Caroline & Music, Bradley & Graczyk, Randolph (eds.), *Papers from CLS 25*, 342–356. Chicago: Chicago Linguistics Society.
- Partee, Barbara & Rooth, Mats. 1983. Generalized conjunction and type ambiguity. In Bäuerle, Rainer & Schwarze, Christoph & von Stechow, Arnim (eds.), *Meaning, use and interpretation of language*, 361–383. Berlin: Walter de Gruyter.
- Patejuk, Agnieszka & Przepiórkowski, Adam. 2023. Category mismatches in coordination vindicated. *Linguistic Inquiry* 54(2). 326–349. DOI: https://doi.org/10.1162/ling_a_00438
- Peterson, Peter. 1981. Problems with constraints on coordination. *Linguistic Analysis* 8(4). 449–460.
- Peterson, Peter. 2004. Coordination: Consequences of a lexical-functional account. *Natural Language and Linguistic Theory* 22. 643–679. DOI: <https://doi.org/10.1023/B:NALA.0000027673.49915.2b>

- Petré, Peter. 2012. General productivity: How *become* waxed and *wax* became a copula. *Cognitive Linguistics* 23(1). 27–65. DOI: <https://doi.org/10.1515/cog-2012-0002>
- Pietroski, Paul. 2007. Systematicity via monadicity. *Croatian Journal of Philosophy* 7. 343–374.
- Pollard, Carl & Sag, Ivan A. 1994. *Head-driven phrase structure grammar*. Chicago: University of Chicago Press.
- Przepiórkowski, Adam. 2022. A compositional intersective account of heterofunctional coordination. In Starr, John R. & Kim, Juhyae & Öney, Burak (eds.), *Proceedings of Semantics and Linguistic Theory* 32 270–293. Lexington, KY: Linguistic Society of America. DOI: <https://doi.org/10.3765/salt.v1i0.5333>
- Przepiórkowski, Adam & Łukasiewicz-Pater, Julia & Kuś, Katarzyna & Maćkiewicz, Bartosz. 2025. Heterofunctional coordination in German. *Journal of Comparative Germanic Linguistics* 28(3). 1–75. DOI: <https://doi.org/10.1007/s10828-024-09159-5>
- Przepiórkowski, Adam & Patejuk, Agnieszka. 2024. Prenominal adverbs, or apparent selectional violations in coordination. *Linguistic Inquiry* Online First. 1–29. DOI: https://doi.org/10.1162/ling_a_00548
- Quine, Willard van Orman. 1956. Quantifiers and propositional attitudes. *Philosophical Review* 53. 177–187. DOI: <https://doi.org/10.2307/2022451>
- Rett, Jessica. 2013. Similitives and the argument structure of verbs. *Natural Language & Linguistic Theory* 31(4). 1101–1137. DOI: <https://doi.org/10.1007/s11049-013-9201-0>
- Ross, John Robert. 1972. Doubl-ing. *Linguistic Inquiry* 3(1). 61–86.
- Sag, Ivan & Gazdar, Gerald & Wasow, Thomas & Weisler, Steven. 1985. Coordination and how to distinguish categories. *Natural Language and Linguistic Theory* 3. 117–171. DOI: <https://doi.org/10.1007/BF00133839>
- Stabler, Edward. 1997. Derivational minimalism. In Retoré, Christian (ed.), *Logical aspects of computational linguistics: First International Conference, LACL '96, Nancy, France, September 1996, selected papers*, 68–95. Berlin: Springer. DOI: <https://doi.org/10.1007/BFb0052152>
- Stabler, Edward. 2011. Computational perspectives on minimalism. In Boeckx, Cedric (ed.), *Oxford handbook of linguistic minimalism*, 617–642. Oxford: Oxford University Press.
- Stabler, Edward. 2013. Two models of minimalist, incremental syntactic analysis. *Topics in Cognitive Science* 5. 611–633. DOI: <https://doi.org/10.1111/tops.12031>
- Zhang, Niina Ning. 2007. The syntactic derivations of two paired dependency constructions. *Lingua* 117. 2134–2158. DOI: <https://doi.org/10.1016/j.lingua.2007.02.001>
- Zhang, Niina Ning. 2010. *Coordination in syntax*. Cambridge: Cambridge University Press.

